

Enumeration and bit fields

Enum

- An enum is a special type that represents a group of constants (unchangeable values)
- To create an enum, use the enum keyword, followed by the name of the enum, and separate the enum items with a comma:
- ```
enum Level {
 LOW,
 MEDIUM,
 HIGH
};
```
- Default value will start from 0. i.e. LOW=0, MEDIUM=1, HIGH=2 likewise

# Cont...

- To access the enum, you must create a variable of it.
- Inside the main() method, specify the enum keyword, followed by the name of the enum (Level) and then the name of the enum variable (myVar in this example):
- `enum Level myVar;`
- ```
int main() {  
    enum Level myVar = MEDIUM;  
    printf("%d", myVar);  
    return 0;  
}
```

Cont...

- enum Level {
 LOW = 5,
 MEDIUM, // Now 6
 HIGH // Now 7
};

- enum Level {
 LOW,
 MEDIUM=6,
 HIGH
};

Cont... Example

- `#include <stdio.h>`
- `enum day {sunday, monday, tuesday, wednesday, thursday, friday, saturday};`
-
- `int main()`
- `{`
- `enum day d = thursday;`
- `printf("The day number stored in d is %d", d);`
- `return 0;`
- `}`
- O/P -> The day number stored in d is 4

bit fields

- struct structure1 {
- int val1;
- int val2;
- };
- struct structure2 {
- int val1: 1;
- int val2: 1;
- };
- int main(void) {
- printf(" \n Size of memory engaged by structure1 : %zu ", sizeof(struct structure1));
- printf(" \n Size of memory engaged by structure2: %zu ", sizeof(struct structure2));
- return 0;
- }